Problem 3

Let us denote the following events:

- B_n: the event that the n-th drawn ball is blue.
- R_n: the event that the n-th drawn ball is red.

The initial probabilities are:

$$\mathbb{P}(B_1) = \frac{b}{b+r}, \quad \mathbb{P}(R_1) = \frac{r}{b+r}$$

Part (a)

The events B_1 and R_1 form a partition of the sample space. Then, applying the Law of Total Probability, we have:

$$\begin{aligned}
\mathbb{P}(B_2) &= \mathbb{P}(B_2 \mid B_1)\mathbb{P}(B_1) + \mathbb{P}(B_2 \mid R_1)\mathbb{P}(R_1) \\
&= \left(\frac{b+d}{b+r+d}\right)\left(\frac{b}{b+r}\right) + \left(\frac{b}{b+r+d}\right)\left(\frac{r}{b+r}\right) \\
&= \frac{b^2+bd+br}{(b+r+d)(b+r)} \\
&= \frac{b(b+r+d)}{(b+r+d)(b+r)} \\
&= \frac{b}{b+r}
\end{aligned}$$

Therefore, the probability that the second drawn ball is blue is:

$$\mathbb{P}(B_2) = \frac{b}{b+r}$$

Analogously, the probability that the second drawn ball is red is:

$$\mathbb{P}(R_2) = \frac{r}{b+r}$$

Part (b)

By Bayes' Formula, we have:

$$\mathbb{P}(B_1 \mid B_2) = \frac{\mathbb{P}(B_2 \mid B_1)\mathbb{P}(B_1)}{\mathbb{P}(B_2)}$$

The conditional probability $\mathbb{P}(B_2 \mid B_1)$ is the probability of drawing a blue ball on the second draw given that the first draw was blue. After drawing a blue ball on the first draw, the urn contains $b+d$ blue balls and r red balls, so:

$$\mathbb{P}(B_2 \mid B_1) = \frac{b+d}{b+r+d}$$

We know that $\mathbb{P}(B_1) = \frac{b}{b+r}$ and, from part (a), $\mathbb{P}(B_2) = \frac{b}{b+r}$. Substituting these values into Bayes' Formula yields:

$$\mathbb{P}(B_1 \mid B_2) = \frac{\frac{b+d}{b+r+d} \cdot \frac{b}{b+r}}{\frac{b}{b+r}} = \frac{b+d}{b+r+d}$$

The probability that the first drawn ball was blue, given that the second is blue, is $\frac{b+d}{b+r+d}$.

Part (c)

Let $\mathbb{P}_{x,y}$ denote the probability measure for an urn initially containing an arbitrary composition of x blue balls and y red balls. By mathematical induction:
Base Case ($n = 1$): By definition, $\mathbb{P}_{b,r}(B_1) = \frac{b}{b+r}$.
Induction Hypothesis: Assume that for a given $n \geq 1$ and for an arbitrary initial composition of x blue balls and y red balls, the probability of drawing a blue ball on the n-th draw is $\mathbb{P}_{x,y}(B_n) = \frac{x}{x+y}$.
Induction Step ($n \to n+1$): We must show that $\mathbb{P}_{b,r}(B_{n+1}) = \frac{b}{b+r}$. As in part (a), the events B_1 and R_1 form a partition of the sample space. Then, applying the Law of Total Probability, we have:

$$\mathbb{P}_{b,r}(B_{n+1}) = \mathbb{P}_{b,r}(B_{n+1} \mid B_1)\mathbb{P}_{b,r}(B_1) + \mathbb{P}_{b,r}(B_{n+1} \mid R_1)\mathbb{P}_{b,r}(R_1) \tag{c.1}$$

Given that B_1 occurs, the urn's composition becomes $b + d$ blue balls and r red balls. The conditional probability defines a new probability measure. Under this measure, the sequence of draws $2, \ldots, n+1$ has the exact same joint distribution as the sequence of draws $1, \ldots, n$ under the measure $\mathbb{P}_{b+d,r}$. Applying the induction hypothesis:

$$\mathbb{P}_{b,r}(B_{n+1} \mid B_1) = \mathbb{P}_{b+d,r}(B_n) = \frac{b+d}{b+r+d}$$

Similarly, if R_1 occurs, the urn updates to b blue balls and $r + d$ red balls:

$$\mathbb{P}_{b,r}(B_{n+1} \mid R_1) = \mathbb{P}_{b,r+d}(B_n) = \frac{b}{b+r+d}$$

Substituting these probabilities, along with the initial probabilities $\mathbb{P}(B_1)$ and $\mathbb{P}(R_1)$, into equation (c.1):

$$\begin{aligned}
\mathbb{P}_{b,r}(B_{n+1}) &= \left(\frac{b+d}{b+r+d}\right)\left(\frac{b}{b+r}\right) + \left(\frac{b}{b+r+d}\right)\left(\frac{r}{b+r}\right) \\
&= \frac{b^2 + bd + br}{(b+r+d)(b+r)} \\
&= \frac{b(b+r+d)}{(b+r+d)(b+r)} \\
&= \frac{b}{b+r}
\end{aligned}$$

Therefore, $\mathbb{P}_{b,r}(B_n) = \mathbb{P}_{b,r}(B_1), \forall n \geq 1$.

$\square$

Part (d)

Let $A_n = B_2 \cap \cdots \cap B_{n+1}$ represent the event that the n consecutive draws following the first draw are blue. We want to find $\mathbb{P}(B_1 \mid A_n)$.

By the definition of conditional probability, we have:

$$\mathbb{P}(B_1 \mid A_n) = \frac{\mathbb{P}(B_1 \cap A_n)}{\mathbb{P}(A_n)} \tag{d.1}$$

Using the Chain Rule (Lemma 2.1.5), the probability of the intersection $B_1 \cap A_n$ (drawing $n+1$

consecutive blue balls) is:

$$\begin{aligned}
\mathbb{P}(B_1 \cap A_n) &= \mathbb{P}(B_1 \cap B_2 \cap \cdots \cap B_{n+1}) \\
&= \mathbb{P}(B_1) \cdot \mathbb{P}(B_2 \mid B_1) \cdot \ldots \cdot \mathbb{P}(B_{n+1} \mid B_1 \cap \cdots \cap B_n) \\
&= \frac{b}{b+r} \cdot \frac{b+d}{b+r+d} \cdots \frac{b+nd}{b+r+nd} \\
&= \frac{\prod_{k=0}^{n}(b+kd)}{\prod_{k=0}^{n}(b+r+kd)} \qquad \text{(d.2)}
\end{aligned}$$

Since the events B_1 and R_1 form a partition of the sample space, the intersections $A_n \cap B_1$ and $A_n \cap R_1$ are disjoint. By σ-additivity, we have:

$$\mathbb{P}(A_n) = \mathbb{P}(A_n \cap B_1) + \mathbb{P}(A_n \cap R_1) \qquad \text{(d.3)}$$

We also have that $\mathbb{P}(A_n \cap B_1) = \mathbb{P}(B_1 \cap A_n)$, which we have already calculated, and:

$$\begin{aligned}
\mathbb{P}(A_n \cap R_1) &= \mathbb{P}(R_1 \cap A_n) \\
&= \mathbb{P}(R_1 \cap B_2 \cap \cdots \cap B_{n+1}) \\
&= \frac{r}{b+r} \cdot \frac{b}{b+r+d} \cdots \frac{b+(n-1)d}{b+r+nd} \\
&= \frac{r\prod_{k=0}^{n-1}(b+kd)}{\prod_{k=0}^{n}(b+r+kd)} \qquad \text{(d.4)}
\end{aligned}$$

Substituting (d.2) and (d.4) into equation (d.3):

$$\begin{aligned}
\mathbb{P}(A_n) &= \frac{\prod_{k=0}^{n}(b+kd)}{\prod_{k=0}^{n}(b+r+kd)} + \frac{r\prod_{k=0}^{n-1}(b+kd)}{\prod_{k=0}^{n}(b+r+kd)} \\
&= \frac{\prod_{k=0}^{n}(b+kd) + r\prod_{k=0}^{n-1}(b+kd)}{\prod_{k=0}^{n}(b+r+kd)} \\
&= \frac{(b+nd)\prod_{k=0}^{n-1}(b+kd) + r\prod_{k=0}^{n-1}(b+kd)}{\prod_{k=0}^{n}(b+r+kd)} \\
&= \frac{(b+nd+r)\prod_{k=0}^{n-1}(b+kd)}{\prod_{k=0}^{n}(b+r+kd)} \qquad \text{(d.5)}
\end{aligned}$$

Substituting (d.2) and (d.5) into equation (d.1):

$$\begin{aligned}
\mathbb{P}(B_1 \mid A_n) &= \frac{\frac{\prod_{k=0}^{n}(b+kd)}{\prod_{k=0}^{n}(b+r+kd)}}{\frac{(b+nd+r)\prod_{k=0}^{n-1}(b+kd)}{\prod_{k=0}^{n}(b+r+kd)}} \\
&= \frac{\prod_{k=0}^{n}(b+kd)}{(b+nd+r)\prod_{k=0}^{n-1}(b+kd)} \\
&= \frac{(b+nd)\prod_{k=0}^{n-1}(b+kd)}{(b+nd+r)\prod_{k=0}^{n-1}(b+kd)} \\
&= \frac{b+nd}{b+r+nd}
\end{aligned}$$

Finally, assuming $d > 0$, we calculate the limit as $n \to \infty$:

$$\lim_{n\to\infty} \mathbb{P}(B_1 \mid A_n) = \lim_{n\to\infty} \frac{b+nd}{b+r+nd} = \lim_{n\to\infty} \frac{\frac{b}{n}+d}{\frac{b+r}{n}+d} = \frac{d}{d} = 1$$

If $d = 0$ (sampling with replacement), then $\mathbb{P}(B_1 \mid A_n) = \frac{b}{b+r}$ for all n (i.e., the events are independent), and the limit as $n \to \infty$ is also $\frac{b}{b+r}$. If $d > 0$, the probability that the first drawn ball was blue, given that the following n balls are all blue, is $\frac{b+nd}{b+r+nd}$, and its limit as $n \to \infty$ is 1.